

HS 312 – Introduction to Science and Technology Studies

Lecture 9

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Risk as an Umbrella Term

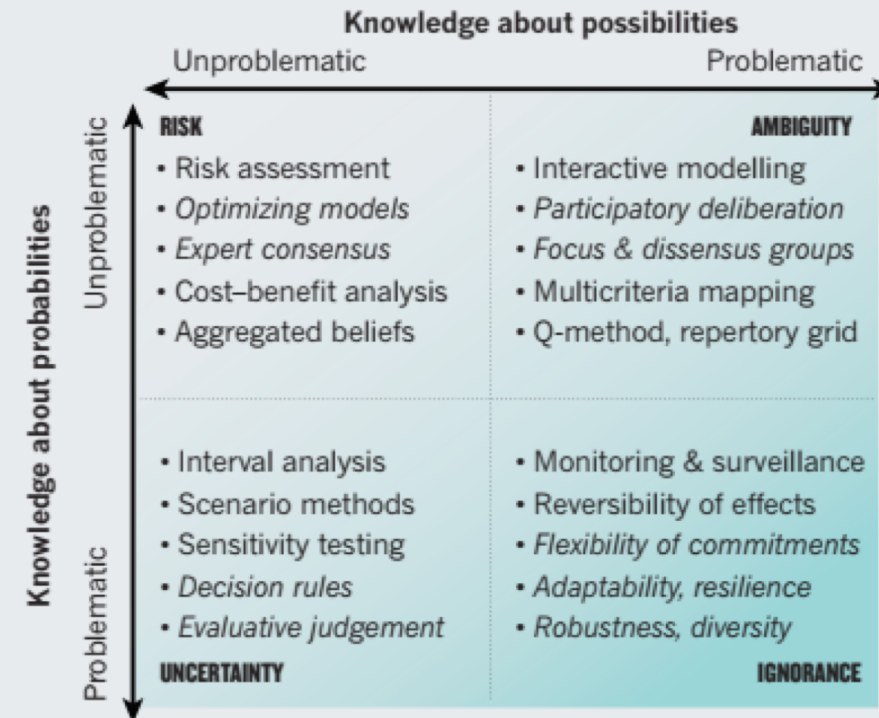
Hazard	Possibility that something (e.g. technology) causes undesirable effects
Risk	Specification of a hazard, usually in terms of probabilities and consequences
Uncertainty	Situation in which hazards are known but cannot be expressed as risks due to a lack of knowledge
Ignorance	Situation in which hazards are unknown
Indeterminacy	Consequences (hazards) are causally open, for example because they depend on user or operator behaviour
Ambiguity	Situation in which there are different disagreeing perspectives on the nature and the (un)desirability of a hazard

Uncertainty Matrix

- Andy Sterling (2010)

UNCERTAINTY MATRIX

A tool to catalyse nuanced deliberations: experts must look beyond risk (top left quadrant) to ambiguity, uncertainty and ignorance using quantitative and *qualitative* methods.



Political pressures tend to push attention from 'plural conditional' (dark shading) to 'single definitive' (light shading) methods.

Sources

- Sismondo Sergio (2010). Questioning Functionalism in the Sociology of Science. Chapter 3, in *An Introduction to Science and Technology Studies*; second edition: Blackwell Publishing, Oxford UK.
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- R.K. Sony, Daniel Münster & Siddhartha Krishnan (2023). *What counts as evidence? Examining the controversy over pesticide exposure and etiology in an environmental justice movement in Kerala, India*, Environmental Sociology, 9:2, 148-164.

Thank you